

Zombie Warfare FPS - Complete Game Design Document

1. Executive Summary

Zombie Warfare FPS is a browser-based multiplayer FPS built with Three.js, TypeScript, Socket.io and MongoDB. The game combines Team Deathmatch and Zombie Mode with weapon progression, skins, shop systems, player levels and competitive gameplay.

2. Core Features

First-person shooter controls, jumping, sprinting, crouching, AK-47, M249, AWP, Knife, multiplayer networking, zombie AI, progression, cosmetics and shop.

3. Technical Stack

Frontend: Three.js, TypeScript, Vite, Zustand. Backend: Node.js, Express, Socket.io. Database: MongoDB Atlas.

4. Gameplay Modes

Team Deathmatch: first team to 50 kills wins. Zombie Mode: players survive infection waves and zombies attempt to infect all humans.

5. Weapon System

AK-47 (balanced), M249 (high ammo), AWP (high damage sniper), Knife (fast melee). Include recoil, reload, spread and headshot multipliers.

6. Player Systems

Health, armor, coins, experience, level progression, statistics tracking and cosmetic customization.

7. Skin System

Weapon skins are divided into Common, Rare, Epic, Legendary and Mythic tiers. Equipped skins are synchronized across clients.

8. Shop System

Players purchase skins, character cosmetics and effects using earned coins. Daily rewards and mission rewards provide currency.

9. Maps

Abandoned Factory, Military Base and Zombie Research Facility. Each map supports multiple game modes.

10. Zombie AI

Finite-state machine: Idle → Patrol → Detect → Chase → Attack → Dead.

11. Networking

Server-authoritative movement validation, hit registration, state synchronization, scoreboards and matchmaking.

12. Database Schema

Collections: users, inventories, skins, matches, missions, leaderboards.

13. UI/UX

HUD, minimap, weapon selection, scoreboard, shop interface, inventory and match-end statistics.

14. Performance Targets

60+ FPS on modern browsers, object pooling, frustum culling, optimized shadows and network compression.

15. Development Roadmap

Phase 1 FPS controller. Phase 2 multiplayer. Phase 3 weapons. Phase 4 progression and shop. Phase 5 ranked systems and content expansion.

Agent Implementation Backlog

- Task 1: Implement feature/module #1 with tests, documentation and integration.
- Task 2: Implement feature/module #2 with tests, documentation and integration.
- Task 3: Implement feature/module #3 with tests, documentation and integration.
- Task 4: Implement feature/module #4 with tests, documentation and integration.
- Task 5: Implement feature/module #5 with tests, documentation and integration.
- Task 6: Implement feature/module #6 with tests, documentation and integration.
- Task 7: Implement feature/module #7 with tests, documentation and integration.
- Task 8: Implement feature/module #8 with tests, documentation and integration.
- Task 9: Implement feature/module #9 with tests, documentation and integration.
- Task 10: Implement feature/module #10 with tests, documentation and integration.
- Task 11: Implement feature/module #11 with tests, documentation and integration.
- Task 12: Implement feature/module #12 with tests, documentation and integration.
- Task 13: Implement feature/module #13 with tests, documentation and integration.
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Task 49: Implement feature/module #49 with tests, documentation and integration.

Task 50: Implement feature/module #50 with tests, documentation and integration.